

The Future Of Database Services: Cloud Database

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Abstract— Databases are the core of all the business services that are offered online. Cloud computing has drastically changed the way these online services are being offered. Hence it is of significance importance that the databases need to be compatible with cloud computing. The data is growing enormously with every passing day. With the paradigm shift in the technology used to store, manage and retrieve the data from file storage to relational databases to today's cloud databases, the need to process the query efficiently while preserving the privacy had always been the priority. The dynamism of the query and the optimization of the resources used for query execution are the aspects that doesn't change in any situation. The timing constraint on the query execution could be either instantaneous, urgent, leisurly or indefinite. But in almost all the scenarios the effective storage, manipulation and data retrieval is the key. While retrieving the data stored in cloud, privacy preservation is of utmost importance. To improve query efficiency in cloud with preserving the privacy can be achieved by introducing dataflow optimization techniques and defining data descriptor based algorithms.

I. INTRODUCTION

Cloud computing technology had changed the way the data is stored and processed. The multidimensional growth in computing systems and technologies have resulted into advanced scalable, portable and large scale integrated systems and technologies. Data centers, virtualization, cloud and WEB2 technologies are the leaders of this. Computing power was earlier based on the products used and usually upgrading was a costly affair. Advances in cloud computing technologies has changed the scene now and computing is being offered as a service over the internet. This has given the industry a relief in terms of capital investment as well as ongoing cost. The private and public cloud platforms offer to deliver all the benefits of cloud computing technology to their customers. Databases had always been the critical part of this endeavour. E.F. Codd in 1970 proposed the rules for relational databases resulting into efficient and optimized query processing. To achieve such effectiveness today's cloud databases need to be made cloud computing compatible. Recent technologies are putting more demand on query optimization and data security with increasing scalability.

The characteristic feature of cloud computing is it provides Infrastructure, Platform and Software as a Service as per the user demand. The demands for these resources are flexible and

the payment model is determined as per the demands without compromising onto the reliability of the services being offered. Following are the major features of the systems implementing cloud with datacenter, virtualization and WEB2 interfaces.

1. Efficiency
2. Fault tolerance
3. Ability to run in heterogeneous environment
4. Ability to operate on encrypted data
5. Ability to interface with business intelligence products

Significance of Resource Allocation in Cloud:

Resource Allocation (RA) is the process of assigning available resources to the needed cloud applications over the internet. The services are no more effective if they are being starved due to inappropriate Resource allocation. Resource provisioning has to be managed such that all the requesting services are being provided available resources effectively.

Resource Allocation Strategy (RAS) designing concentrates on the following factors:

- 1) availability of the scarce resources
 - 2) utilization of the resources by the service providers
 - 3) type of resources needed
 - 4) time for which a particular resource will be held to complete a user job.
 - 5) sequence in which resources will be held
- It is expected that the optimal RSA will have the best solution for the following situations:
- 1) decision to be made when a same resource is being requested by two applications at the same time instance.
 - 2) when limited resources are available
 - 3) how an isolated resource could be allocated as needed
 - 4) in case of surplus availability of the resources, which to be allocated
 - 5) in case of scarcity of resources, which will be assigned

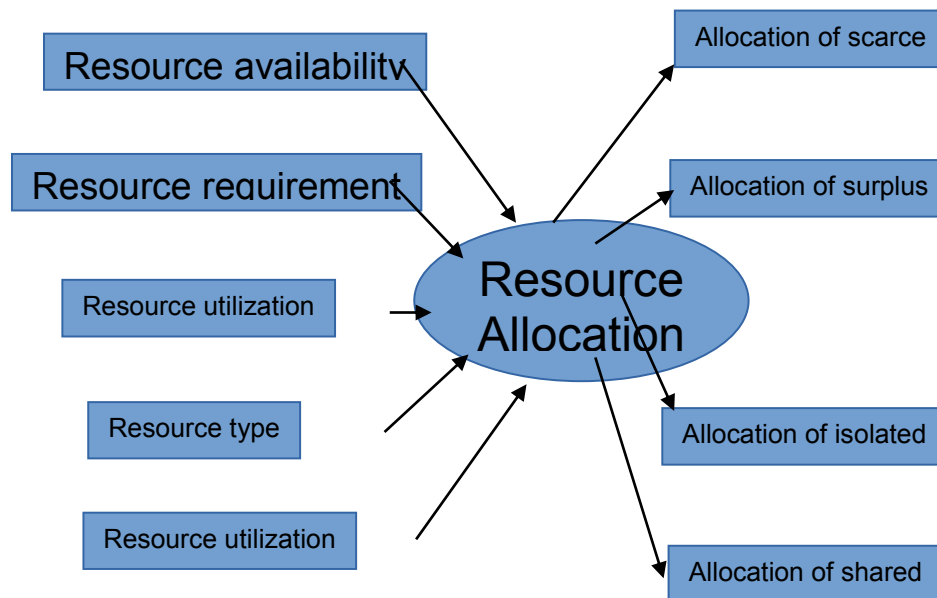


Fig. 1. Resource allocation strategy

Service level agreement (SLA) is the document that defines the criteria for service availability to the user. An optimal resource allocation strategy will always try to find out the solution to the situation wherein the scarce resources too are being made available to the user on demand as guaranteed in SLA. As the user demands are flexible and difficult to predict beforehand, it's the crucial task for the cloud service provider to judge these demands and make the resources available. On the other hand, the cloud user is concerned about the time taken by the service provider to complete the assigned task with minimal cost. Hence limited resources, heterogeneous nature of resources, varying user and application demands pose the challenges for the RAS. Virtualization technology advancements make physical provisioning of the resources possible.

Advantages of resource allocation

- 1) without bothering about the hardware infrastructure, installed software, user can develop and deploy the application with ease over the internet.
- 2) the application is scalable and available at minimal cost
- 3) resource sharing is practically possible in case of the scarcity of the resources.

Limitation of resource allocation

- 1) cloud service providers have the autonomy over the shared resources.
- 2) cloud service user faces the vendor lock in problem and transferring huge data across different service providers is difficult.
- 3) data privacy policies at public cloud are insufficient

Privacy issues in cloud:

Information security and data privacy are the terms used extensively interchangeably in the industry. It is assumed that the information security policies mentions and maintains the data privacy as well. But in the cloud perspective, though these terms are interrelated, privacy needs to be taken care of explicitly. The concept of privacy is very relative and changes very often among the countries, cultures, jurisdiction and the organizational structures.

The data passes through the following phases and the privacy needs to be maintained at every stage.

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7
Generation	Use	Transfer	Transformation	Storage	Archival	Destruction

When the data is passing through all these phases, the organization needs to keep checking for the following:

- 1) how the personal information will be shared with the individual requesting it through different platforms?
- 2) when the data is moved to the cloud, how the individual's personal information will be kept private across the migration?
- 3) if the cloud service provider migrates the data from one storage location to another without individual's consent, how the privacy will be guaranteed in the changed legislation?
- 4) what is the lifespan of the data storage for personal and sensitive information?
- 5) data destruction needs to ensure that the data is not just inaccessible but is completely destroyed.

Challenges in existing system:

We will study the challenges in the existing system with reference to the Eucalyptus. The Walrus storage provides the

list of private data to the users. Client's access request for public or protected data can be satisfied on demand. Walrus storage has buckets as a container defined to store the data. The Eucalyptus Walrus architecture is as shown:

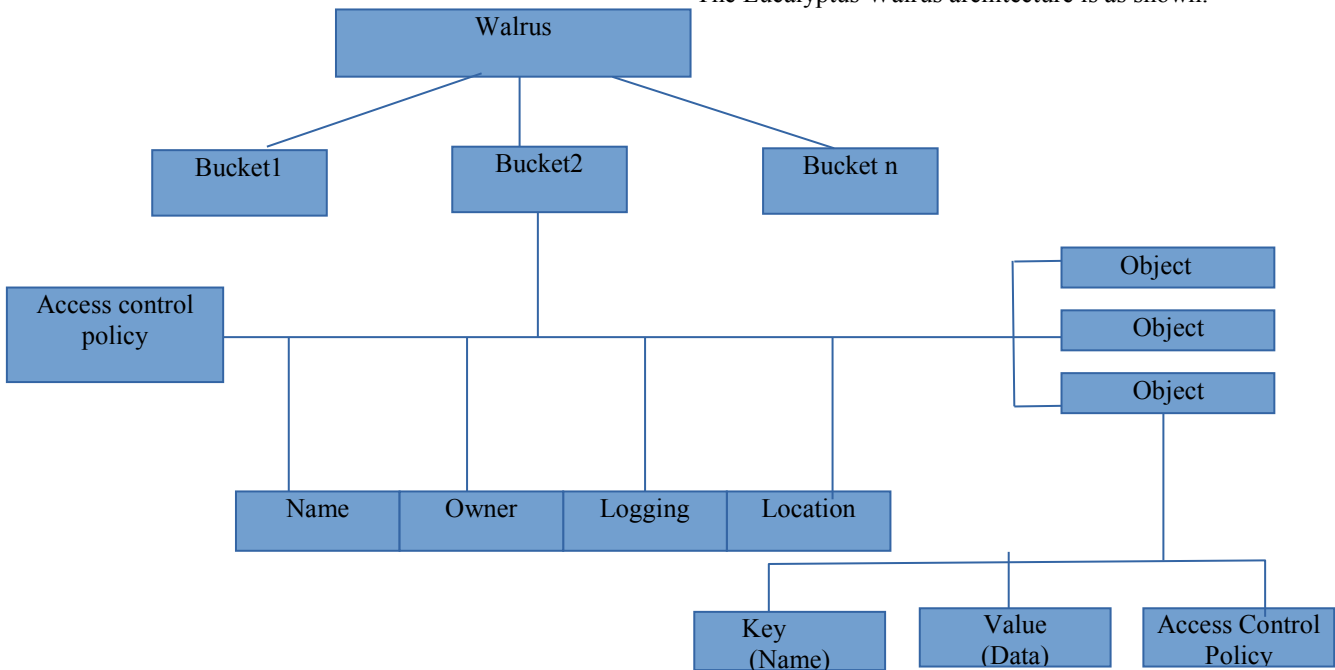


Fig. 2. Eucalyptus Walrus Architecture

But client need to specify the bucket name or the access URL for the bucket in advance. Usually the time required to locate the private bucket is less than the time required to access public or protected bucket because access control list of the bucket needs to be searched to check whether or not the user can access it. And revoking the access policy to reduce the search time for the buckets may compromise the privacy of the buckets.

In a cloud, access control policies are not defined by the data owner or provider. It is being decided by the architecture used to maintain the confidentiality as well as the performance. Whenever a user is requesting for any data as a result of the query, it needs to establish the identity to the server. So this is an overhead to many advantages of storing the data at external sites. Hence it is very much desired that there is some mechanism that can let the data enforce the restrictions onto the set of users who can access it.

The malicious users who are keeping track of the user requests being generated at the server may get the user personal information as well as the access to the private data during the cloud service provider's interaction with the user. The user's sensitive information may be compromised with to in turn affect the desired query results. The query execution is largely dependent on to the integrity of the data. The result to be correct it must be operated on to the genuine data, to be complete the whole data collection be made available on request and to be fresh the recent data be made available. Digital signatures, defined authenticated data structures which

are flexible to accommodate data can guarantee correctness and completeness. Freshness can be provided by making authenticated data structures dependent on a variable that changes over time.

II. METHODOLOGY

We will study a hypothetical situation wherein a cloud service user has developed an application. This is a web based serach service which the client wants to host on to the cloud platform. This service will be made available remotely and is expected to deal with the scalability issue effectively. As the load on the application increases, so will increase the demand for the resources needed by the client. It is expected that the client can acquire and release the resources dynamically and inexpensively. This poses a challenge for the cloud service provider.

Especially when the data services are the compositions of the other possibly third party services (eg. Google search or Yahoo Image search), these services becomes the data processing graphs that use the third party services as building blocks and invoke them during their execution. Running these data services under different QoS (Quality of Service) constraints as per the clients requirements further makes the system complex.

To decide upon which third party services to be used in processing correctly, making the optimal use of the resources available, satisfy all the QoS constraints is quite difficult. To make data services scalable and adaptable to the cloud

environment, the dataflow needs to be optimized automatically. This is analogues to query optimization and execution in traditional databases.

The basic building blocks of any query optimization algorithm are:

1. Query Evaluation Plan generation
2. Search strategy
3. Cost function.

Cloud services are expected to provide the personal data to the user on demand on varied platforms. This indeed makes this sensitive data vulnerable to security and privacy. Every time a client is accessing the cloud based service, his/her personal credentials are exposed to the world if the data privacy is not being ensured. Cloud computing is a world of heterogeneous data and services. When such data is being accessed, the keyword based queries plays an important role to ease out the query processing. Developing index structures to support querying hybrid data is difficult. With emerging new technologies having potential to create new data management scenarios in which users join ad-hoc communities to create, collaborate, curate and discuss data online. Data virtualization is the presentation of data as an abstract layer, independent of underlying database systems, structures and storage. Database virtualization is the decoupling of the database layer which lies between the storage and application layers within the application stack.

Herein we propose to use the Eucalyptus private cloud architecture as the experimental setup for testing the effectiveness of privacy preservation strategy in storage clouds. Storage clouds allows the client to upload the data to the servers. The data is available as and when required with the implemented security policies and reliability. The most difficult task for the storage clouds is to identify the authorized user for the requested data from the huge collection. Cloud providers allow for direct access to the private data. It is expected that only the authorized clients from the list provided to get access to the private data be allowed and others be denied the service by the data provider.

Let S be a system that stores the data in the cloud such that:

$$S = \{i, c, e, n, d \mid F(d), F(n), F(e)\}$$

where

i = index for data storage

c = capacity of the data storage device

e = access specifier (public, private or protected) for the stored data

n = node at which the data is stored

d = data present on cloud

for every $i \in S$

$$F(d) : i \rightarrow d$$

$$F(n) : i \rightarrow n$$

$$F(e) : i \rightarrow e$$

Let U be the system of cloud service users such that

$$U = \{u, I_{\text{private}}, I_{\text{protected}}, I_{\text{public}} \mid F(\text{private}), F(\text{protected}), F(\text{public})\}$$

where

$$u = \text{user}$$

$$I_{\text{private}} = \text{private data}$$

$$I_{\text{protected}} = \text{protected data}$$

$$I_{\text{public}} = \text{public data}$$

for every user $u \in U$

$$F(\text{private}) : u \times i \rightarrow I_{\text{private}}$$

$$F(\text{protected}) : u \times i \rightarrow I_{\text{protected}}$$

$$F(\text{public}) : u \times i \rightarrow I_{\text{public}}$$

We propose the system for data storage organization which will be way more effective by reorganizing the data based on the access specifications provided by the cloud service user. This system will also ensure the privacy of the data being intact. This will be achieved by using the descriptor based data organization.

III. CONCLUSION

In the existing system, the data access policies are not effective for public or protected data retrieval. Data confidentiality has to be balanced with query processing functions and performance. Most cloud platforms fail to support SQL queries as they are not designed to support structured data management. For maintaining the data privacy, storage structures could be modified to have self-imposed privacy measures. To improve query efficiency in cloud with preserving the privacy can be achieved by introducing dataflow optimization techniques and defining data descriptor based algorithms.

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